

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I **JAIVANT.S (192311326)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **JAIVANT.S**

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis

Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

ANSWERS

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications. Analyze Performance, Security, and Manageability, then Justify the Better Choice. (10 Marks)

Introduction

A **hypervisor** is virtualization software that allows multiple virtual machines (VMs) to run on a single physical server. Hypervisors are classified into **Type-1 (Bare-Metal)** and **Type-2 (Hosted)** based on how they are deployed. Choosing the right hypervisor is critical for mission-critical cloud applications where high availability, security, and performance are essential.

Type-1 Hypervisor

- Installed directly on the physical hardware without a host operating system.
- Provides direct access to CPU, memory, storage, and networking resources.
- Examples: VMware ESXi, Microsoft Hyper-V, Xen.

Advantages

- High performance due to minimal overhead.
- Better resource utilization.
- Strong security because of a smaller attack surface.
- Supports advanced features such as live migration, fault tolerance, clustering, and high availability.

Type-2 Hypervisor

- Installed on top of a host operating system.

- The host OS manages hardware resources before they reach the virtual machines.
- Examples: Oracle VirtualBox, VMware Workstation.

Advantages

- Easy to install and configure.
- Suitable for software testing, development, and education.
- Supports multiple guest operating systems on personal computers.

Limitations

- Lower performance because requests pass through the host OS.
- Higher resource consumption.
- Greater security risks due to dependence on the host operating system.

Comparison

Performance

Type-1 hypervisors provide near-native performance with low latency and efficient CPU scheduling. Type-2 hypervisors introduce additional overhead, making them slower under heavy workloads.

Security

Type-1 hypervisors offer stronger isolation between virtual machines and a reduced attack surface. Type-2 hypervisors inherit vulnerabilities from the host operating system, making them less secure.

Manageability

Type-1 hypervisors support centralized management, live migration, snapshots, resource pooling, and disaster recovery. Type-2 hypervisors provide limited enterprise management features.

Justification

Mission-critical applications require high uptime, fast response, strong security, and efficient resource management. Therefore, **Type-1 hypervisors are the preferred choice** because they deliver higher performance, better isolation, improved scalability, and enterprise-grade management capabilities.

Conclusion

For cloud datacenters hosting banking, healthcare, government, or enterprise applications, **Type-1 hypervisors are the ideal solution** due to their superior performance, security, reliability, and manageability.

2. A Cloud Provider Must Isolate Workloads Belonging to Finance, Healthcare, and Student Portals on the Same Hardware. Analyze How Virtualization Architecture Supports Isolation and Identify Two Major Attack Surfaces. (10 Marks)

Introduction

Cloud providers often host multiple organizations on the same physical infrastructure. Virtualization enables **multi-tenancy**, allowing different workloads to share hardware while maintaining complete isolation and security.

Virtualization Architecture

A hypervisor creates multiple virtual machines (VMs), each with its own operating system, applications, CPU allocation, memory, storage, and network interface. Every workload runs independently without interfering with others.

For example:

- Finance applications run in one VM.
- Healthcare applications run in another VM.
- Student portal applications run in a separate VM.

Each VM has dedicated virtual resources, ensuring secure separation.

How Isolation is Achieved

CPU Isolation

The hypervisor schedules processor time fairly among all VMs, preventing one workload from monopolizing CPU resources.

Memory Isolation

Each VM receives dedicated memory space. Memory protection prevents one VM from reading or modifying another VM's data.

Storage Isolation

Virtual disks are separated so that one organization's data cannot be accessed by another.

Network Isolation

Virtual LANs (VLANs) and Software Defined Networking (SDN) isolate network traffic between workloads.

Major Attack Surfaces

1. Hypervisor Attack

If attackers exploit vulnerabilities in the hypervisor, they may gain control over multiple VMs hosted on the server.

2. VM Escape Attack

A compromised virtual machine escapes the virtual environment and accesses the host operating system or neighboring VMs.

Security Measures

- Regular hypervisor updates
- Multi-factor authentication
- Network segmentation
- Encryption of stored and transmitted data
- Continuous monitoring and intrusion detection

Conclusion

Virtualization architecture enables secure workload isolation through resource separation and access control. Although hypervisor attacks and VM escape are major threats, proper security practices significantly reduce these risks.

3. Study the Following VM Host Utilization Snapshot and Recommend Corrective Actions: CPU 92%, Memory 88%, Storage I/O Latency 35 ms, Average VM Boot Time 170 s. Explain the Likely Bottlenecks. (10 Marks)

Introduction

Monitoring VM performance helps identify resource bottlenecks that affect cloud application performance and availability.

Given Data

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O Latency = 35 ms
- Average VM Boot Time = 170 seconds

Analysis

CPU Bottleneck

CPU utilization above 90% indicates processor saturation. Applications may experience slower execution and increased response times.

Memory Bottleneck

Memory utilization of 88% is close to full capacity. The host may begin swapping memory to disk, reducing performance.

Storage Bottleneck

A storage latency of 35 ms is relatively high, indicating slow disk access. This affects application responsiveness and VM startup time.

Long Boot Time

A VM boot time of 170 seconds suggests that CPU overload and storage delays are slowing the initialization process.

Corrective Actions

Improve CPU Performance

- Add more processor cores.
- Distribute workloads across additional hosts.
- Enable CPU load balancing.

Increase Memory

- Upgrade RAM.
- Optimize memory allocation.
- Remove unused virtual machines.

Improve Storage

- Replace HDDs with SSD or NVMe drives.
- Use storage tiering and caching.
- Optimize disk scheduling.

Optimize Virtual Machines

- Remove unnecessary startup services.
- Use VM templates.
- Enable live migration to balance workloads.

Continuous Monitoring

Use monitoring tools to track CPU, memory, storage, and network usage in real time.

Conclusion

The major bottlenecks are CPU overload, high memory usage, and storage latency. Upgrading hardware, balancing workloads, and optimizing storage will significantly improve VM performance.

4. Analyze How Software Defined Datacenter (SDDC) Architecture Improves Agility Compared with a Traditional Datacenter. Support Your Answer with Compute, Storage, and Network Virtualization Considerations. (10 Marks)

Introduction

A **Software Defined Datacenter (SDDC)** is a modern datacenter where compute, storage, and networking resources are fully virtualized and managed through software. Unlike traditional datacenters that rely on manual hardware configuration, SDDCs provide automation, flexibility, and scalability.

Compute Virtualization

- Multiple virtual machines share physical servers.
- Resources such as CPU and memory are allocated dynamically.
- Supports live migration, high availability, and fault tolerance.
- Improves server utilization and reduces hardware costs.

Storage Virtualization

- Combines multiple physical storage devices into a single logical storage pool.
- Storage can be allocated on demand.
- Supports snapshots, replication, and disaster recovery.
- Simplifies storage management.

Network Virtualization

- Creates virtual networks independent of physical hardware.
- Uses Software Defined Networking (SDN) for centralized management.
- Enables quick deployment of virtual networks.
- Improves network security through segmentation and isolation.

Advantages Over Traditional Datacenters

- Faster deployment of applications.
- Automated resource provisioning.
- Better scalability for changing workloads.
- Reduced operational costs.
- Improved disaster recovery and business continuity.
- Centralized monitoring and management.

Conclusion

SDDC architecture significantly improves agility by virtualizing compute, storage, and networking resources. It enables automation, faster provisioning, higher scalability, and better resource utilization, making it ideal for modern cloud environments.

5. A Multi-Cloud Federation Project Fails Due to Identity Mismatches Between Providers. Analyze the Issue and Propose a Secure Identity and Privacy Design Using Federation Concepts. (10 Marks)

Introduction

In a multi-cloud environment, organizations use services from multiple cloud providers. Identity federation allows users to access resources across different clouds using a single digital identity.

Problem Analysis

The project fails because different cloud providers maintain separate identity management systems. This results in:

- Duplicate user accounts.
- Authentication failures.
- Inconsistent user permissions.
- Difficult identity management.
- Increased security risks.

Secure Identity Design

Single Sign-On (SSO)

Allows users to authenticate once and access multiple cloud services without repeated logins.

Identity Federation

Uses trusted Identity Providers (IdPs) and standards such as:

- SAML

- OAuth 2.0
- OpenID Connect

These standards allow secure exchange of authentication information.

Multi-Factor Authentication (MFA)

Requires additional verification such as OTPs or biometric authentication to improve security.

Role-Based Access Control (RBAC)

Users receive permissions based on their job roles, reducing unauthorized access.

Privacy Protection

- Encrypt identity information.
- Use secure communication protocols (TLS).
- Maintain audit logs.
- Share only the minimum required user information between cloud providers.

Benefits

- Improved security
- Better user experience
- Simplified identity management
- Regulatory compliance
- Reduced administrative overhead

Conclusion

Identity federation, combined with SSO, MFA, RBAC, encryption, and privacy protection, provides secure authentication across multiple cloud providers and ensures seamless access while protecting sensitive user information.
